

Day 1 — Mathematics

Comprehensive Problem Set

Wednesday, June 17, 2026 • 18 questions

Today's setup first: create or log into your MyNCEES account and download the FE Reference Handbook (the exact version is on the NCEES website). Confirm your calculator is on the NCEES-approved list. Then do this set with the handbook open.

How to take it: Give yourself about 50–55 minutes (FE pace is ≈ 3 min/question). Answer everything — no penalty for guessing. Then check the answer key and study every worked solution, noting the handbook section used. Everything here lives in the handbook's *Mathematics* section.

Questions

- Solve $2x^2 - 5x - 3 = 0$.
(A) $x = 3, -\frac{1}{2}$ (B) $x = -3, \frac{1}{2}$
(C) $x = 2, -3$ (D) $x = 1, \frac{3}{2}$
- If $3^x = 81$, then $x =$
(A) 3 (B) 4
(C) 27 (D) 3.3
- Solve the system $2x + 3y = 12$ and $x - y = 1$.
(A) (3, 2) (B) (2, 3)
(C) (4, 1) (D) (1, 4)
- In a 3–4–5 right triangle, the angle opposite the side of length 3 is closest to:
(A) 30° (B) 36.9°
(C) 45° (D) 53.1°
- Evaluate $\sin 30^\circ + \cos 60^\circ$.
(A) 0.50 (B) 0.87
(C) 1.00 (D) 1.50
- The distance between points (1, 2) and (4, 6) is:
(A) 3 (B) 4
(C) 5 (D) 7
- A line passes through (0, 1) and (2, 5). Its equation is:
(A) $y = 2x + 1$ (B) $y = x + 1$
(C) $y = 2x - 1$ (D) $y = 3x + 1$
- The equation $(x - 2)^2 + (y + 1)^2 = 25$ is a circle with center and radius:
(A) center (2, -1), $r = 25$ (B) center (-2, 1), $r = 5$
(C) center (2, -1), $r = 5$ (D) center (-2, 1), $r = 25$
- For $f(x) = 3x^3 - 2x^2 + x$, the derivative $f'(2) =$
(A) 17 (B) 23
(C) 29 (D) 35

10. Evaluate $\int_0^4 (2x + 3) dx$.
- (A) 20 (B) 24
(C) 28 (D) 32
11. Evaluate $\int_1^3 \frac{1}{x} dx$.
- (A) 0.48 (B) 1.10
(C) 2.00 (D) 3.00
12. The function $f(x) = x^2 - 6x + 5$ has a minimum value of:
- (A) -4 (B) 0
(C) 3 (D) 5
13. A contaminant decays by first order, $\frac{dC}{dt} = -kC$, with $k = 0.10 \text{ day}^{-1}$. Its half-life is closest to:
- (A) 5.0 days (B) 6.9 days
(C) 10 days (D) 14 days
14. Using the trapezoidal rule with two equal intervals ($h = 1$), estimate $\int_0^2 x^2 dx$.
- (A) 2.0 (B) 2.67
(C) 3.0 (D) 4.0
15. The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is:
- (A) 0 (B) 1
(C) ∞ (D) undefined
16. The dot product $(2, 3, 1) \cdot (1, -1, 4) =$
- (A) 0 (B) 3
(C) 7 (D) 9
17. The determinant $\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} =$
- (A) 5 (B) 8
(C) 11 (D) -5
18. The Newton–Raphson method is used to:
- (A) integrate a function numerically (B) find roots of an equation iteratively
(C) solve linear systems by elimination (D) perform least-squares regression

Answer Key

1	2	3	4	5	6	7	8	9
A	B	A	B	C	C	A	C	C
10	11	12	13	14	15	16	17	18
C	B	A	B	C	B	B	A	B

Worked Solutions

- Answer: A.** Quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{25 + 24}}{4} = \frac{5 \pm 7}{4} = 3$ or -0.5 .
Handbook: Mathematics — Quadratic Equation.
- Answer: B.** $81 = 3^4$, so $x = 4$. Or $x = \frac{\log 81}{\log 3} = 4$. Handbook: Mathematics — Logarithms.
- Answer: A.** From the second equation $x = 1 + y$. Substitute: $2(1 + y) + 3y = 12 \Rightarrow 2 + 5y = 12 \Rightarrow y = 2$, $x = 3$.
Handbook: Algebra — simultaneous equations.
- Answer: B.** The angle opposite the side of length 3 has $\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}$, so $\theta = \arctan(0.75) = 36.9^\circ$.
Handbook: Trigonometry — right-triangle relations.
- Answer: C.** $\sin 30^\circ = 0.5$ and $\cos 60^\circ = 0.5$, so the sum = 1.0. Handbook: Trigonometry — common-angle values.
- Answer: C.** $d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Handbook: Analytic Geometry — distance formula.
- Answer: A.** Slope $m = \frac{5 - 1}{2 - 0} = 2$; y -intercept $b = 1$, so $y = 2x + 1$. Handbook: Analytic Geometry — straight line.
- Answer: C.** Standard form $(x - h)^2 + (y - k)^2 = r^2$ gives center $(2, -1)$ and $r = \sqrt{25} = 5$.
Handbook: Analytic Geometry — conic sections (circle).
- Answer: C.** $f'(x) = 9x^2 - 4x + 1$; $f'(2) = 36 - 8 + 1 = 29$. Handbook: Differential Calculus — polynomial derivatives.
- Answer: C.** $\int (2x + 3) dx = x^2 + 3x$; evaluate $0 \rightarrow 4$: $(16 + 12) - 0 = 28$. Handbook: Integral Calculus.
- Answer: B.** $\int \frac{1}{x} dx = \ln x$; $\ln 3 - \ln 1 = 1.099 \approx 1.10$. Handbook: Integral Calculus — standard integrals.
- Answer: A.** Vertex at $x = -\frac{b}{2a} = \frac{6}{2} = 3$; $f(3) = 9 - 18 + 5 = -4$ (a minimum, since $a > 0$).
Handbook: Differential Calculus — maxima/minima.
- Answer: B.** First-order half-life $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.10} = 6.93$ days. This same equation drives BOD decay, disinfection, and radioactive decay. Handbook: Mathematics — differential equations / decay.

14. **Answer: C.** Trapezoidal rule: $\frac{h}{2}[f(0) + 2f(1) + f(2)] = \frac{1}{2}[0 + 2(1) + 4] = 3.0$. (Exact value is $\frac{8}{3} = 2.67$; the rule overestimates a convex curve.) Handbook: Numerical Methods — numerical integration.
15. **Answer: B.** A standard limit: $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$. Handbook: Differential Calculus — L'Hôpital's rule / limits.
16. **Answer: B.** Dot product = $2(1) + 3(-1) + 1(4) = 2 - 3 + 4 = 3$. Handbook: Vectors — dot (scalar) product.
17. **Answer: A.** 2×2 determinant = $(2)(4) - (1)(3) = 8 - 3 = 5$. Handbook: Matrices — determinants.
18. **Answer: B.** Newton–Raphson iterates $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ to find roots. Handbook: Numerical Methods — root finding.

Score yourself: Tally how many you missed and in which subtopic (algebra, trig, geometry, calculus, numerical). Bring any weak subtopic into tomorrow's warm-up. **Tomorrow (Day 2): Fluid Mechanics I.**